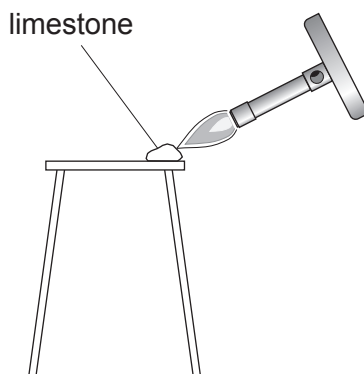
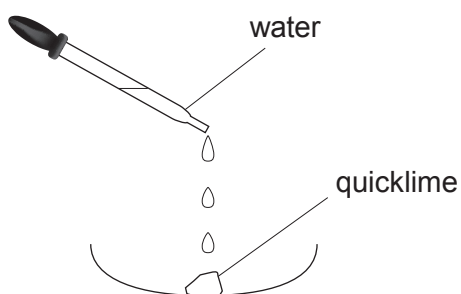


5. (a) A student carried out a two-stage experiment to change limestone (calcium carbonate) into slaked lime (calcium hydroxide).

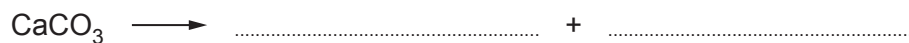


Stage 1: Limestone (calcium carbonate) decomposes into quicklime (calcium oxide) and carbon dioxide



Stage 2: Quicklime (calcium oxide) reacts with water forming slaked lime (calcium hydroxide)

- (i) Write the formulae for calcium oxide and carbon dioxide to complete the equation for the reaction taking place in stage 1. [2]



- (ii) Calcium hydroxide contains one  $\text{Ca}^{2+}$  ion for every two  $\text{OH}^-$  ions.

Write the chemical formula for calcium hydroxide. [1]

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A black and white photograph of a large-scale industrial quarry or processing plant. In the center, a complex of buildings and machinery is visible, with a thick plume of white smoke or steam rising into the air. The facility is situated in a valley, flanked by steep, rocky cliffs. The foreground is dominated by large, light-colored piles of material, likely crushed stone or sand, with several heavy trucks and loaders operating on the ground. The background shows more of the quarry's infrastructure and the surrounding landscape, including a road and some vegetation on the upper slopes.

